

Data Analytics Final Project Guidelines

Project Objective

The final project aims to assess your ability to perform comprehensive data analysis using Python. You are expected to demonstrate end-to-end exploratory data analysis (EDA), starting from data cleaning to deriving insights through visualizations and statistical summaries.

Project Scope

Your project will involve the following key components:

1. Data Loading and Initial Overview

- *Import the dataset using Pandas and provide an overview:*
 - Number of rows and columns
 - Data types of each column
 - Initial observations (e.g., `head()`, `info()`, `describe()`)

2. Data Pre-processing

- *Perform all necessary cleaning steps such as:*
 - Handling missing values
 - Removing duplicates
 - Correcting data types
 - Creating derived columns
 - Filtering or aggregating data

3. Exploratory Data Analysis (EDA)

- *Conduct descriptive and exploratory analysis to uncover patterns and trends:*
 - Univariate, bivariate, and multivariate analysis
 - Use groupby, pivot tables, and correlation analysis
 - Include statistical summaries to support findings

4. Visualizations

- *Use **Matplotlib** / **Seaborn** / **Plotly** to generate meaningful visualizations:*
 - Bar plots, line charts, pie charts, histograms, box plots, scatter plots, heatmaps, etc
 - Ensure visuals have proper titles, labels, legends, and color schemes

- Use subplots where applicable for a better layout

5. Insight Generation and Report

- *Summarize key insights from your analysis:*
 - Use Markdown cells throughout the notebook to clearly explain your logic, methods, and interpretation of results
 - Highlight significant patterns, correlations, or anomalies found in the data
 - Conclude with a short summary of your overall findings and any potential recommendations or next steps

Project Domains

Here are some sample domain ideas to help you get started.
However, you are free to choose any domain or topic of your interest.

1. Sales & E-commerce

- Sales performance dashboard
- Customer segmentation based on region or product category
- Trend and seasonality analysis of sales data

2. Finance

- Expense tracking dashboard
- Budget vs. Actual analysis
- Profitability by product or region

3. Healthcare

- Hospital/clinic patient records analysis
- Disease count trends across months/years
- Treatment cost analysis

4. Retail

- Inventory and stock analysis
- Customer purchasing patterns
- Store-wise performance comparison

5. Education

- Student performance dashboard

- Course enrollment and completion analysis
- Feedback and satisfaction trends
- 6. Human Resources (HR)**
 - Employee attrition or retention reports
 - Department-wise performance or salary comparison
 - Leave and attendance trends

7. Environmental

- Traffic pattern analysis across regions or time periods
- Air quality monitoring and pollutant trends
- Water consumption or waste generation tracking

8. Social Media & Public Sentiment

- Sentiment analysis of social media posts or movie reviews (e.g., Twitter, IMDB) ○
- Trending hashtags or topics analysis
- Public opinion analysis on a campaign or event

Dataset Requirements

The dataset should have:

- **Minimum of 500 records**
- **At least 10 features/columns** (preferably a mix of numerical and categorical data)

Data Sources

Here are some reliable sources where you can find datasets for your project:

1. **UCI Machine Learning Repository:** [UCI Repository](#)
2. **Google Dataset Search:** [Google Dataset Search](#)
3. **Data.gov (U.S.):** [Data.gov](#)
4. **Data.gov.in (India):** [Data.gov.in](#)
5. **World Health Organization (WHO):** [WHO Data](#)
6. **India Data Portal:** [India Data Portal](#)

7. **Awesome Public Datasets:** [Awesome Public Datasets](#)
8. **OpenML:** [OpenML](#)
9. **FiveThirtyEight:** [FiveThirtyEight Datasets](#)
10. **Hugging Face** <https://huggingface.co/datasets>
11. <https://indiadataportal.com/>
12. learn.microsoft.com
13. <https://universe.roboflow.com/>

Submission Requirements

1. Original Dataset Source

- The raw data file in **CSV** or **Excel** format (as obtained from the source)

2. Jupyter/Colab Notebook `.ipynb`

- A well-structured and well-commented notebook containing:
 - ★ **Complete Python code** covering data loading, cleaning, analysis, and visualizations
 - ★ **At least 10 different visualizations** using `matplotlib` / `seaborn` / `plotly`
 - ★ **Explanatory Markdown cells** throughout the notebook describing each step, decision, and interpretation
 - ★ **At least 5 key insights** derived from the data and explained in your own words

Data Analytics Final Project Rubric [Total: 50 Marks]

TASK 1: Problem Definition & Dataset Selection (10 marks)

Clear problem statement, relevant domain, quality dataset (size, source, variety).

TASK 2: Data Cleaning & Pre-processing (10 marks)

Handling missing data, duplicates, formatting, derived features, and transformation.

TASK 3: EDA & Visualizations (15 marks)

Depth of analysis (univariate, bivariate, multivariate) with 10+ visuals using `matplotlib`, `seaborn`, and/or `plotly`.

TASK 4: Documentation, Insights & Presentation (15 marks)

Well-commented notebook, clear insights (at least 5), summaries, and clean presentation.